

Tutorial:Basic SSTO Design

This tutorial explains how to build a single-stage-to-orbit (SSTO) spaceplane that is able to launch into space and return without jettisoning any parts.

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Powerplant

Any craft that hopes to get anywhere needs engines. For SSTO craft there are two popular options: jet engine/liquid fuel engine combinations and the CR-7 R.A.P.I.E.R.

Jet Engines

Jet engines are high-efficiency, high-thrust, low-cost engines but require an atmosphere to operate. In the case of SSTO craft, we will need jet engines that can perform as high as possible to justify their inability to perform in a vacuum. Currently the J-X4 "Whiplash" Turbo Ramjet Engine is by far the most sensible choice. While the J-33 "Wheesley" Turbofan Engine is lower cost and consumes less fuel and money, the Whiplash is capable of decent performance at high altitudes where the Wheesley cannot keep up.

Jet Engines are powered by liquid fuel and intake air. Liquid fuel can be supplied from regular rocket tanks, but jet fuselages are a far more sensible choice due to their lower cost and mass per unit of liquid fuel. Intake air is provided by air intakes. Depending on how many you place, your jet

engines' effective service ceiling can be anywhere from 15km to 30km. The Shock Cone Intake is currently the air intake with the highest amount of potential air intake.

Liquid Fuel

Liquid fuel engines, which burn both liquid fuel and oxidizer, are what will carry your craft into orbit once the air becomes too thin for jet engines to operate. The T-1 Toroidal Aerospike "Dart" Liquid Fuel Engine is one of the better choices due to its low profile and high specific impulse (it performs better than other engines). For more economy-minded builders, the LV-T45 "Swivel" Liquid Fuel Engine is not a bad choice. However, its huge size can make it tricky to take off from the runway without destroying the engine.

R.A.P.I.E.R Engines

The CR-7 R.A.P.I.E.R. Engine, introduced in KSP version 0.23, is your go-to "two in one" solution for all your SSTO needs. It can function as both an air breathing engine in "Air Breathing" mode and a much more powerful rocket engine in "Closed Cycle" mode. Putting out 465.462 kN of thrust in Kerbin's atmosphere at mach 3.7 and 180 kN in vacuum, the CR-7 R.A.P.I.E.R engine is truly a beast of an SSTO engine. Although the engine can automatically switch modes, for best performance one should watch the thrust output (by right clicking the engine in flight) and forcibly switch all rapiers to closed cycle once your orbital speed plateaus. Setting up an action group is the best way to do this and to close the intakes at the same time.

Stability

One of the most important factors to how well an SSTO will perform is its aerial stability. A stable craft is going to be more desirable than an unstable craft. There are several forces in Kerbal Space Program which have an effect on the flight characteristics of SSTO craft.

Center of mass

One is Center of mass. This is the point from which there is an equal amount of mass in each direction and acts as a fulcrum for any rotational forces. The center of mass of any craft will shift as a result of burning fuel, but this is especially critical for SSTOs as spin-outs and otherwise catastrophic stability failures will hinder its ability to reach space.

Center of lift

The second is Center of lift. This is arguably the most important characteristic an SSTO designer should consider when building. To prevent craft spin-outs and otherwise catastrophic stability failures, it is important that the center of lift always be to the rear of the center of mass.

Center of thrust

A sometimes overlooked factor is Center of thrust. This is where force will be acting upon your craft to push it forward. Though this is a non-issue for most SSTOs, it is wise to check that your center of thrust is roughly in line with your center of mass and lift.

The bottom line: Ensure your center of mass is always ahead of your center of lift and your center of thrust is in line with your center of mass.

Additional Systems

These are common features of SSTO craft. The command pod is, as always, a must, and the others, though technically optional, are recommended in most cases.

Command Pod

The possibilities are limitless. You can create either manned or unmanned SSTOs. Just remember than manned ones have crew and unmanned have electricity. The Mk1 Cockpit is an example of a manned command pod, while the Probodobodyne HECS is an unmanned command pod. A probe will require a fairing or cargo bay to maintain aerodynamics.

Takeoff Gear

It is possible to launch a craft using the TT18-A Launch Stability Enhancer and land using parachutes. For conventional SSTOs, however, the LY-10 Small Landing Gear is what you'll use to take off. Remember to retract these wheels right after takeoff to reduce drag, as well as to extend right before landing.

A parachute can be useful for SSTOs as it removes the need to perform a landing. Simply deploy to return to the ground safely.

SAS

Since the standard cockpits are supplied with their own SAS, it is mostly unnecessary to add additional SAS (such as reaction wheels) to your SSTO. However, large craft may require external SAS to help maneuver with the extra mass, and you can always disable external SAS at your whim.

SAS is a stability augmentation system and helps your craft fly straight. In order to accomplish this, it provides torque (the rotational analogue of force). This helps your craft change directions. Control surfaces can provide torque while moving in atmospheres, while reaction wheels can provide torque while stationary or in vacuum.

RCS

A Reaction Control System is required for maneuvering in a vacuum, a must-have if you plan to actually perform tasks in orbit. Simple orbiting/deorbiting does not require RCS, though an RCS coupled with SAS can make flying heavier/less stable craft much easier.

RCS is activated the with the "R" button with stock controls, first and foremost. RCS is fueled exclusively by monopropellant tanks. When your monopropellant stores are depleted, your RCS will fail to function.

Aerodynamics

Specifically Wings and Control Surfaces. These are what allow the SSTO to function as an atmospheric craft. The Delta Wing is a solid choice, with a large surface area handy for attaching additional objects to, as well as being rather symmetrical and easy to work with.

However, the Delta Wing is fixed and cannot provide any kind of maneuvering (it only makes your plane fly). In order to turn in-atmosphere, you are going to need control surfaces. The numerous elevons, rotating winglets, and canards are all considered control surfaces. Control surfaces have the distinction of being to pivot or otherwise react to controls, providing lift in certain directions. This is contrasted with regular winglets, which are fixed and cannot move.

Power

Depending on your conditions, you may need to provide additional electric charge for your SSTO. This can be stored in batteries and/or supplied by solar panels, RTGs, or fuel cells. Electricity is used to turn the ship via its reaction wheels, to transmit science back to Kerbin with antennas, to conduct research in Mobile Processing Labs, to operate ion engines, and to operate mining excavators and resource converters. Without sufficient electric charge, crafts may be unable to rotate due to loss of power, and unmanned spacecraft may become completely inoperable. In general, at least 1-2 solar panels are recommended for nearly everything going into space.

Extra batteries may be desirable if you do not have enough power production to meet the needs of your antenna(s) when transmitting data back to Kerbin. For example, if you're going to take an ion-powered spaceplane to Eeloo and want to use a Communotron 88-88 antenna to send back an orbital gravity scan, the antenna will require 100 electricity per second for 3 seconds. You could store that much energy in three Z-100 Rechargeable Battery Packs (0.03 mass), or you could try to produce it as you use it with 125 RTGs (10 mass), or, due to the distance from Kerbol, you'd require 244 Gigantor XL Solar Arrays (73.2 mass). The batteries are the obvious winner in this case.

Putting the pieces together

Now that the basics of SSTO design have been covered, it is time for you to assemble your first SSTO.

Remember to consider everything above to create a working SSTO.

Troubleshooting

Answers likely questions people may run into when designing SSTOs. However, if you even try to build an SSTO you will need a basic understanding of Kerbal Space Program physics, so check there as well if this can't answer your question(s).

It doesn't take off.

Make sure your landing gear is placed such that your craft can achieve a positive angle of attack while it's on the runway. Typically you'll have either a nose wheel and two wheels slightly behind centre of mass that act as a fulcrum (your elevators push down on the tail and lift the nose, rotating around the rear wheels), or a tail wheel that's placed higher than two other wheels (which are ahead of the centre of mass) so that the craft has a positive angle of attack even when stationary on the runway and will naturally lift off at the right speed without pitching upwards.

It spins out on the runway.

Make sure your landing gear is placed symmetrically and your center of mass and lift are both in the center of the craft. A common issue with wing mounted landing gear is they will be 5 degrees toe in or 10 degrees toe out. Use the SHIFT key to make smaller rotations to get them aligned straight. You will also need horizontal aerodynamic stability, which is classically accomplished with a vertical tail fin. As with launch stability of rockets, you may find your pilot has too much control authority in the yaw dimension and is overcompensating - try using a different part as a tail fin.

I keep blowing up my engines when trying to lift off

Be slightly gentler on the stick, change the layout of your engine modules to be slightly forwards, or make your landing gear taller so your tail won't hit the ground when you pitch up.

It flips out whenever I move/start flying.

This is likely a result of a poor center of mass/lift. If this occurs just after liftoff, check these in the [Spaceplane Hangar](#); you will be most likely find your answer there. If this occurs late-flight, it is a result of either something breaking off or fuel moving around and changing the center of mass (probably the latter). This is a relatively easy to fix problem. Before entering the atmosphere again, move your fuel forwards to hopefully place the center of mass ahead of the center of lift.

I can't reach orbit

This is the result of not having enough Delta-V, or just poor flying. For most SSTOs, a safe procedure is to: Take off and climb at only 5 degrees. This should allow the RAPIERs to accelerate past 400m/s (the speed at which their thrust begins to increase) easily. Slowly pitch up to avoid overheating. Keep at around 15 degrees to allow the plane to accelerate past 1000m/s. At around 20-25km, your engines will flame out and so you should switch to your vacuum engines. Cut the engine when your Ap passes 70km. Wait until you get near to your Ap and circularise.

I hit the runway when landing and died.

Coming in for hot landings (usually above 80 m/s) will cause your aircraft to either bounce off the runway or blow up. Slow your descent by throttling back when around a kilometer from the runway, then nose up slightly and deploy landing gear and flaps (right click your ailerons and elevators and deploy them - this will generate extra lift and drag). When you contact with the tarmac, apply brakes. If you die again you were either too fast or your center of mass was off. If you can't slow down enough, try adding air brakes and deploying them before landing.

How can I make landing easier?

Just burn retrograde when in orbit until your trajectory is intersecting with the planet. Have a parachute or two on standby and deploy them when you think you're close enough to the ground (they will activate for you but will induce significant G-force on your craft). You will probably not land where you wanted.

Is there any way to make it better than it already is?

Chances are if you have already reached this point it is unnecessary to create a new craft for the purpose of this previous craft. Your SSTO is no longer a prototype and is finally a complete, but better yet working design! You can always modify the design to suit a particular need.

See also

- [Tutorial: Basic Plane Design](#)
- [Tutorial: Spaceplane basics](#)
- [Tutorial:Spaceplane to Laythe](#)

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